

ZIHAO ZHANG, PH.D.

Designer · Educator · Scholar in Landscape Architecture

City College of New York, Spitzer School of Architecture

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researchgate.net/profile/Zihao-Zhang-10

SUMMARY

Zihao Zhang is a designer, educator, and scholar in landscape architecture. He studies landscape as a product of technology and labor, asking what machines, tools, and unseen work of care produce the environments we call natural, and how that care is distributed among humans and nonhumans, including intelligent machines. He develops theory through making, prototyping with sensors, actuators, and fabrication. His book *Cybernetics and the Constructed Environment* (Routledge, 2025) critiques the cybernetic legacy in design, which, grounded in Western conceptions of nature, treats environments as systems to be sensed, modeled, and controlled. It proposes instead a design practice that begins where the model ends, among more-than-human agents that resist capture. His current research turns to Chinese thought and landscapes as a distinct cosmology of technology, nature, and cultivation, which he extends in his next book, *Solarpunk Landscapes: Towards Radical Sustainability* (with Shurui Zhang, under contract with Routledge). He is a co-founder of OSE (Office of Speculative Ecologies), a studio working across climate adaptation, urban ecology, and social justice.

HIGHER EDUCATION

Ph.D. in the Constructed Environment, University of Virginia, School of Architecture, 2021

Dissertation: *Cybernetic Environment: Uncontrollability and Non-communication for a Future of Coexistence*

Committee: Michael Lee, Nana Last, Bradley Cantrell, Michael Gorman

Master of Landscape Architecture, University of Virginia, School of Architecture, 2016

Bachelor of Engineering in Landscape Architecture, Beijing Forestry University, 2013

EXPERIENCE

Leadership

2024–2026	Director, Master of Landscape Architecture Program, Spitzer School of Architecture, CCNY
2023–2024	Interim Director, Master of Landscape Architecture Program, Spitzer School of Architecture, CCNY
2023–2025	Co-chair, ASLA-NY Climate Action Committee

Academic Appointments

2026–present	Associate Professor, City College of New York
2021–2026	Assistant Professor (tenure-track), City College of New York
2019–2021	Lecturer, Landscape Architecture, University of Virginia
2017–2019	Teaching Fellow, Landscape Architecture, University of Virginia
2015, 2016	Graduate Instructor, Summer Design Institute, University of Virginia

Professional Experience

2023–present	Co-founder, OSE: Office of Speculative Ecologies
2014	Landscape Design Intern, Waterstreet Studio, Charlottesville, VA
2013	Landscape Design Intern, Tsinghua Urban Planning and Design Institute, Beijing
2012	Architectural and Urban Design Intern, China Construction Engineering Design Group, Beijing

ACADEMIC AND PROFESSIONAL HONORS

June 2024	Journal of Digital Landscape Architecture Award for Scientific Excellence, Vienna
May 2023	Journal of Digital Landscape Architecture Award for Scientific Excellence, Bernburg
April 2019	Finalist, University Graduate Teaching Awards, University of Virginia
June 2017	Journal of Digital Landscape Architecture Award for Scientific Excellence, Bernburg
2014	VA-ASLA Award for Community Service

RESEARCH, SCHOLARSHIP, AND CREATIVE WORK

Books

(Under contract) Zhang, Zihao, and Shurui Zhang. *Solarpunk Landscapes: Towards Radical Sustainability*. New York: Routledge.

Zhang, Zihao. *Cybernetics and the Constructed Environment: Design Between Nature and Technology*. New York: Routledge, 2025. <https://doi.org/10.4324/9781003320852>

Book Chapters

(Forthcoming) Zhang, Zihao. "Do Machines Care? Posthuman Labor in the Design Professions." In *The Routledge Companion to Artificial Intelligence in Architecture*, 2nd ed., edited by Imdat As. New York: Routledge, 2027.

(Forthcoming) Zhang, Zihao, and Shurui Zhang. "Escaping Western Gaze: Chinese Diasporic Urbanism in Flushing." In *Becoming Chinatown: An Anthology of Interiors*, edited by Leyuan Li and Tairan An. Berlin: Jovis, 2027.

(Forthcoming) Zhang, Zihao, and Shurui Zhang. "One Prompt, Job Done: A Syndrome." In *Representing Landscapes: AI, Machine Vision, and Climate Futures*, edited by Nadia Amoroso. New York: Routledge, 2027.

Macwillie, Elizabeth, Shawn Rickenbacker, and **Zihao Zhang**. "Strengthening Local Ecologies of Care Through Art and Activism in Harlem, NYC." In *Activism in Design Education: Health, Equity, and Climate Action*, edited by Claire Latane and Jean Yang, 95–108. New York: Routledge, 2027.

Zhang, Zihao, and Shurui Zhang. "Before the After: Representing Climate Actions in the Age of AI." In *Representing Landscapes: Visualizing Climate Action*, edited by Nadia Amoroso. Routledge, 2024.
<https://doi.org/10.4324/9781003404798> (Peer-reviewed; Major contribution)

Cantrell, Bradley, **Zihao Zhang**, and Xun Liu. "Artificial intelligence and machine learning in landscape architecture." In *The Routledge Companion to Artificial Intelligence in Architecture*, edited by Imdat As & Prithwish Basu. Taylor & Francis, 2021. (Peer-reviewed; Equal contribution)

Gorman, Michael E., **Zihao Zhang**, et al. "Collaboration Among Apparently Incommensurable Expertises." In *The Third Wave in Science and Technology Studies*, 255–72. Springer, 2019. (Peer-reviewed; Major contribution)

Journal Articles

Fernberg, Phillip, and **Zihao Zhang**. "Problematizing AI Omnipresence in Landscape Architecture." *Journal of Digital Landscape Architecture* 9 (2024): 746–55. <https://doi.org/10.14627/537752069> (Double-blind peer-reviewed)

Zhang, Zihao, et al. "Robots in the Garden: Artificial Intelligence and Adaptive Landscapes." *Journal of Digital Landscape Architecture* 8 (2023): 264–72. (Double-blind peer-reviewed; Lead author)

Zhang, Zihao, and Shurui Zhang. "Mapping Landscape Architects' Expertise in Climate Adaptation." *Landscape Architecture Frontiers* 10.2 (2022): 71–81. (Double-blind peer-reviewed; Lead author)

Zhang, Zihao. "Post-Digital Landscape and Post-Digital Culture." *Journal of Digital Landscape Architecture* 7 (2022): 26–35. (Double-blind peer-reviewed)

Zhang, Zihao, and Bradley Cantrell. "Cultivated Wildness: Towards Technodiversity and Wildness in Machines." *Landscape Architecture Frontiers* 9.1 (2021): 52–65. (Peer-reviewed; Lead author)

Zhang, Zihao, and Xun Liu. "Control and Uncertainty: Towards a Paradigm of Prototyping." *Landscape Architecture Frontiers* 8.4 (2020): 10–25. (Double-blind peer-reviewed; Lead author)

Zhang, Zihao. "Cybernetic Environment: A Historical Reflection on System, Design, and Machine Intelligence." *Journal of Digital Landscape Architecture* 5 (2020). (Double-blind peer-reviewed)

Zhang, Zihao, and Ben Bowes. "The Future of AI and ML in Landscape Design." *Journal of Digital Landscape Architecture* 4 (2019): 2–9. (Double-blind peer-reviewed; Lead author)

Cantrell, Bradley, and **Zihao Zhang**. "A Third Intelligence." *Landscape Architecture Frontiers* 6.2 (2018): 42–51. (Peer-reviewed; Equal authorship)

Zhang, Zihao, Andrew Mondschein, and Mona El Khafif. "Human-Tool Assemblage: Designers in the Big Data World." *Journal of Digital Landscape Architecture* 2 (2018): 397–405. (Double-blind peer-reviewed; Lead author)

Zhang, Zihao. "Human Factors in Responsive Landscapes: Importance and Method." *Journal of Digital Landscape Architecture* 2 (2017): 10–17. (Double-blind peer-reviewed)

Manuscripts Under Review

Zhang, Zihao. "Jugaad Epistemology: Community Sensing and the Knowledge That Lives in the Gap." *Landscape Architecture Frontiers*, special issue on Sustainable Landscape Architecture Supported by Pioneering Sensing Technology. Under review, 2026.

Conference Proceedings

Zhang, Zihao. "On Laboratories and Machines: Interfacing with Landscape Processes." *CELA 2023 Annual Conference Proceedings*. (Double-blind peer-reviewed)

Karkut, Nicholas, Alexey Kiriluk, **Zihao Zhang**, Zhigang Zhu. "Vegetation Coverage and Urban Amenity Mapping Using Computer Vision and Machine Learning." *International Conference on Image Processing and Vision Engineering*, 2023. (Double-blind peer-reviewed)

Chen, G.Q., N. Kashyap, **Z. Zhang**, et al. "Predicting Building Energy Efficiency with Multi-Task Learning." *International Conference on SMART MULTIMEDIA*, 2024. (Double-blind peer-reviewed)

Cantrell, Bradley, and **Zihao Zhang**. "Choreographing Intelligent Agents." *107th ACSA Annual Meeting Proceedings*, 2018.

Book Reviews

Zhang, Zihao. "The Dawn of Everything by David Graeber and David Wengrow." *Journal of Landscape Architecture* 2–2024: 88–89. <https://doi.org/10.1080/18626033.2024.2466341>

Essays and Op-eds

Zhang, Shurui, and **Zihao Zhang.** "Resilient Schist: Crafting Harlem's Corridor of Art, Culture, and Ecology." *New York Amsterdam News*, January 30, 2025.

Zhang, Zihao. "AI in Landscape Architecture: Beyond the Myth of AI-Human Rivalry." *Landezine*, January 6, 2025. <https://landezine.com/ai-in-landscape-architecture-beyond-the-myth-of-ai-human-rivalry/>

Selected Presentations and Talks

2025

- "Do Machines Care? Another Solarpunk Manifesto." University of Massachusetts Amherst. October 2, 2025
- Invited Panelist. Digital Landscape Architecture Conference, Dessau (virtual). June 4, 2025
- "Do Machines Care?" Beyond Maintenance Colloquium, ETH Zürich. February 28, 2025

2024

- Invited lecture. University of Pennsylvania Stuart Weitzman School of Design. February 28, 2024
- Invited lecture. "Of Bicycles and Artificial Intelligence." CCNY/CWE. March 25, 2024
- Invited lecture. University of Colorado Boulder. September 10, 2024

2023

- Invited webinar. Landscape Architecture Foundation. August 29, 2023 (1000+ live attendees, LAF record)
- Invited panelist. ASLA 2023 Annual Conference, Minneapolis. October 29, 2023
- Invited panelist. Bi-City Biennale of Urbanism\Architecture (UABB), Shenzhen. January 7, 2023

2022–2018 (selected)

- "Post-Digital Landscape and Post-Digital Culture." Digital Landscape Architecture Conference, Harvard GSD, 2022
- "Cybernetic Environment." Digital Landscape Architecture Conference (virtual), 2020
- Invited respondent for N. Katherine Hayles' lecture, UVA, 2020
- Digital Landscape Architecture Conferences: Dessau 2019, Freising 2018, Bernburg 2017

Other Scholarly and Creative Work

2023–present	Laboratory for Living Systems , NYC. Co-director
2022	Ecological Laboratory for Urban Agriculture (ELUA) , NYC. Designer, PI. https://ce-lab.one/ELUA
2020	Ostenda illuminata — Public installation, Charlottesville, VA
2018	CyborgBloom — Digital Art Installation, Charlottesville, VA
2018	"Forming Atmosphere" — Landscape design research featured in <i>Codify</i> (Routledge)

Public Scholarship

Chubby Cat Happy Chat Podcast (2024–present). Co-host with Shurui Zhang. Mandarin Chinese podcast on design, philosophy, art, and technology. <https://chubbycathappychat.podbean.com/>

GRANTS, FELLOWSHIPS, AND AWARDS

Grants

2024–2027	DOE Climate Lighthouse. Co-PI. \$1,000,000.
2024–2025	Architectural League Independent Projects Grant. PI (with Shurui Zhang). \$10,000.
2024	CUNY GRTI Round 24. PI. \$9,538.67.
2022–2025	CUNY Energizing Equity. Co-PI. \$600,000. <i>(ended)</i>
2022–2023	CUNY Interdisciplinary Research Grant — Synergistic Agriculture. PI. \$40,000. <i>(ended)</i>
2023–2024	PSC-CUNY — Robots in the Garden. PI. \$6,000. <i>(ended)</i>
2022–2024	PSC-CUNY — Rooftop Thermodynamics. PI. \$6,000. <i>(ended)</i>
2022	Spitzer Faculty Book Project Award. \$4,000.

Fellowships

2023–2024	CUNY Career Success Fellow
2022–2023	Deans' Equity and Inclusion Initiative (DEII) Fellow, Dumbarton Oaks
2020–2021	Graduate Teaching Fellow (\$24,000), University of Virginia
2017–2018	Digital Art Fellow, University of Virginia
2016–2019	PhD Scholarship with full tuition + stipend (\$25,000/yr), University of Virginia

PROFESSIONAL ACTIVITIES

Boards, Workshops, and Committees

2025–present	Executive Editorial Board, <i>Landscape Architecture Frontier</i>
2024–present	Advisory Board, West Harlem Art Fund
2023–2026	Board member (ex officio), ASLA-NY
2023–present	Editorial Board, <i>Journal of Digital Landscape Architecture</i>
2023–2024	Editorial Board, <i>PLOS ONE</i>

Funding Proposal Reviewer

- BARD — US-Israel Agricultural Research & Development Fund (2025)
- Austrian Science Fund (2024)
- European Commission's European Research Council (2023)

Consulting

2023–present Landscape consultant, Masaryk Towers (Mitchell-Lama affordable housing), Lower East Side, NYC

Membership

- American Society of Landscape Architects, New York Chapter (2021–present)

INSTRUCTIONAL ACTIVITIES

City College of New York

Fall 2024	LAAR 63100/65100 — <i>Technological Sublime in a Watery City</i> (6 cr)
Spring 2024	SUS 7501 — <i>Climate Justice in NYC</i> (1 cr)
Fall 2023	LAAR 63100/65100 — <i>Technologic</i> (6 cr)
Spring 2023	ARCH/LAAR Topics — <i>Machines and Ecologies</i> (3 cr)
Fall 2022	LAAR 61100 — <i>Contaminated Diversity</i> (6 cr)
Spring 2022	LAAR 62100 — <i>Cultivated Wildness: Decentralize Central Park</i> (6 cr)
Fall 2021	LAAR 61100 — <i>Cultivated Wildness: Decentralize Central Park</i> (6 cr)
Fall 2021	LAAR 61400 — Digital and Traditional Drawing (3 cr)

University of Virginia

Fall 2020	SARC/LAR6710 — Design Computation (Co-instructor)
Fall 2019	LAR6010 — MLA Foundation Studio (Section instructor)
Spring 2019	LAR7020 — Foundation Studio 4 (Teaching Fellow)
Spring 2018	LAR 8020 — Landscape Manifesto Studio (Teaching Fellow)

UNIVERSITY AND DEPARTMENTAL SERVICE

Committees

- Executive Committee, Sustainability in the Urban Environment Master Program (2025–present)
- Personnel and Budget; Curriculum; Graduate Programs Steering Committees, Dept. of Architecture (2023–present)
- Course and Standing Committee, Dept. of Architecture (2022–present)
- Graduate Program Admission Committee, Dept. of Architecture (2021–present)

Search Committees

- Chair, Associate/Full Professor of Architecture and Urban AI, Dept. of Architecture (2026)
- Lecturer, Sustainability MUE Program; Lab Technician, Robotics Lab; Assistant Professor of LA (2026)
- Professor and Director of MLA Program; Assistant Professor of LA (2024)
- Assistant Director, Bond Center for Urban Futures; Faculty Secretary, Dept. of Architecture (2022–2023)

COMMUNITY OUTREACH

- West Harlem Art Fund and Brotherhood Sister Sol — climate resilience advocacy (2024–present)
- ELUA and SSA rooftop garden tours; partnerships with NYC Composting Project and CCNY Rangel Initiative (2023)
- GrowNYC and NYC-GreenThumb community outreach as part of Synergistic Agriculture project (2021–2022)
- Community-centered Urban Sensing outreach, Charlottesville, VA (2018)

GRADUATE RESEARCH EXPERIENCE

University of Virginia, School of Architecture

- MainStreet21 – Research Coordination Network (PI: Ila Berman, NSF), 2018
- Lhasa Virtual Reality Project (PI: David Germano), 2013–2016
- Winneba Lagoon project, Ghana (PI: Nancy Takahashi), 2015–2016
- James River Viewshed Studies (PI: Guoping Huang), 2013–2014

University of Virginia, School of Engineering

- dMIST – Data-driven Management for Interdependent Stormwater and Transportation Systems (PI: Jonathan Goodall, NSF), 2018–2021